§4.2.

P1. solution:

(a) $b=\begin{bmatrix}1\\2\\2\end{bmatrix}$, $a=\begin{bmatrix}1\\1\\1\end{bmatrix}$

$$p=\left(\frac{a^Tb}{a^Ta}\right)a=\frac{5}{3}\begin{bmatrix}1\\1\\1\end{bmatrix}=\begin{bmatrix}\frac{5}{3}\\\frac{5}{3}\\\frac{5}{3}\end{bmatrix}$$

$$e=b-p=\begin{bmatrix}1\\2\\2\end{bmatrix}-\begin{bmatrix}5/3\\5/3\\5/3\end{bmatrix}=\begin{bmatrix}-2/3\\1/3\\1/3\end{bmatrix}$$

(b) $b=\begin{bmatrix}1\\3\\1\end{bmatrix}$, $a=\begin{bmatrix}-1\\-3\\-1\end{bmatrix}$

$$p=\left(\frac{a^Tb}{a^Ta}\right)a=\left(\frac{-11}{11}\right)\begin{bmatrix}-1\\-3\\-1\end{bmatrix}=\begin{bmatrix}1\\3\\1\end{bmatrix}$$

$$e=b-p=0.$$

P3. Solution:

(a) $$P=\frac{aa^T}{a^Ta}=\frac{1}{3}\begin{bmatrix}1&1&1\\1&1&1\\1&1&1\end{bmatrix}$$

(b) $$P=\frac{aa^T}{a^Ta}=\frac{1}{11}\begin{bmatrix}1&3&1\\3&9&3\\1&3&1\end{bmatrix}$$

$\underline{P^2=P}$

P6. Solution:

$a_1=(-1,2,2)$

$$P_1=\frac{a_1a_1^T}{a_1^Ta_1}=\frac{1}{9}\begin{bmatrix}1&-2&-2\\-2&4&4\\-2&4&4\end{bmatrix}$$

$a_2=(2,2,-1)$

$$P_2=\frac{a_2a_2^T}{a_2^Ta_2}=\frac{1}{9}\begin{bmatrix}4&4&-2\\4&4&-2\\-2&-2&1\end{bmatrix}$$

$a_3=(2,-1,2)$

$$P_3=\frac{a_3a_3^T}{a_3^Ta_3}=\frac{1}{9}\begin{bmatrix}4&-2&4\\-2&1&-2\\4&-2&4\end{bmatrix}$$

$$P_1+P_2+P_3=I$$

P9. Solution:

$P=A(A^TA)^{-1}A^T$ where $A=\begin{bmatrix}1&1\\0&2\end{bmatrix}$

$=I$ since A is invertible.

P16. Solution:

$A=\begin{bmatrix}1&1\\2&0\\-1&1\end{bmatrix}$, $b=\begin{bmatrix}2\\1\\1\end{bmatrix}$

$$A^TA=\begin{bmatrix}6&0\\0&2\end{bmatrix}$$

$$A^Tb=\begin{bmatrix}3\\3\end{bmatrix}$$

$$A^TA\hat{x}=A^Tb\Rightarrow\hat{x}=\begin{bmatrix}\frac{1}{2}\\\frac{3}{2}\end{bmatrix}$$

$$\Rightarrow p=A\hat{x}=\begin{bmatrix}7/6\\1\\1/3\end{bmatrix}$$

P18. Solution:

(a) <u>the line through $(-1,1)$</u>

(b) <u>the Plane perpendicular to the line through $(1,1,1)$</u>

P21. Solution: <u>itself.</u>

P30. Solution:

(a) $P_C=I$. (deleting last column from A).

(b). $A^*=A^T=\begin{bmatrix}3&4\\6&8\\6&8\end{bmatrix}$, linearly dependent.

$$\rightarrow\tilde{A}^*=\begin{bmatrix}3\\6\\6\end{bmatrix}$$

$$\Rightarrow P_R=\frac{\begin{bmatrix}3\\6\\6\end{bmatrix}[3\ 6\ 6]}{[3\ 6\ 6]\begin{bmatrix}3\\6\\6\end{bmatrix}}=\begin{bmatrix}9&18&18\\18&36&36\\18&36&36\end{bmatrix}\cdot\frac{1}{81}$$

$$=\frac{1}{9}\begin{bmatrix}1&2&2\\2&4&4\\2&4&4\end{bmatrix}$$

$$B=P_CAP_R=\frac{1}{9}\begin{bmatrix}27&54&54\\36&72&72\end{bmatrix}=\begin{bmatrix}3&6&6\\4&8&8\end{bmatrix}=A.$$

§4.3.

P1. Solution:

$$A=\begin{bmatrix}1&0\\1&1\\1&3\\1&4\end{bmatrix}\quad b=\begin{bmatrix}0\\8\\8\\20\end{bmatrix}$$

$$A^TA=\begin{bmatrix}4&8\\8&26\end{bmatrix}\quad A^Tb=\begin{bmatrix}36\\112\end{bmatrix}$$

Solve

$$\begin{bmatrix}4&8\\8&26\end{bmatrix}\begin{bmatrix}C\\D\end{bmatrix}=\begin{bmatrix}36\\112\end{bmatrix}$$

$$\Rightarrow \begin{cases}C=1\\D=4\end{cases}$$

$$P_i=C+Dt_i \Rightarrow P=\begin{bmatrix}1\\5\\13\\17\end{bmatrix}$$

$$e=b-P=\begin{bmatrix}-1\\3\\-5\\3\end{bmatrix}$$

$$E=e_1^2+e_2^2+e_3^2+e_4^2=44$$

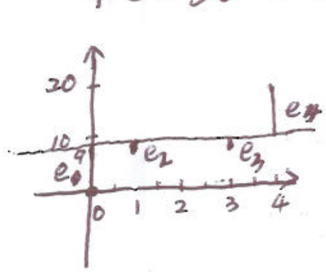
P5. Solution: $y=C$.

$$A=\begin{bmatrix}1\\1\\1\\1\end{bmatrix},\quad b=\begin{bmatrix}0\\8\\8\\20\end{bmatrix}$$

$$A^TA=4,\quad A^Tb=36$$

Solve

$$4C=36 \Rightarrow C=9.$$



$$e_1=-9,\ e_2=-1,\ e_3=-1,\ e_4=11$$

P6. solution:

$$\hat{x}=\frac{a^Tb}{a^Ta}=\frac{36}{4}=9$$

$$P=\hat{x}a=(9,9,9,9)$$

$$e=b-P=(-9,-1,-1,11)$$

$$\|e\|=\sqrt{(-9)^2+(-1)^2+(-1)^2+11^2}=\sqrt{204}=2\sqrt{51}$$

P11. Solution:

$$y=1+4t \rightarrow 1+4(2)=9.$$

Write

$y=C+D(t-\hat{t})$, then $A=\begin{bmatrix}1&t_1-\hat{t}\\\vdots&\vdots\\1&t_m-\hat{t}\end{bmatrix}$

$$A^TA=\begin{bmatrix}m&0\\0&\sum_{i=1}^{m}(t_i-\hat{t})^2\end{bmatrix}$$

$$A^Tb=\begin{bmatrix}\sum_{i=1}^{m}b_i\\\sum_{i=1}^{m}(t_i-\hat{t})b_i\end{bmatrix}=\begin{bmatrix}m\hat{b}\\\sum_{i=1}^{m}(t_i-\hat{t})b_i\end{bmatrix}$$

$\Rightarrow$ The first equation of $A^TA\hat{x}=A^Tb$ is

$$mC=m\hat{b} \Rightarrow C=\hat{b}$$

$\Rightarrow (\hat{t},\hat{b})$ is on the best line $y=\hat{b}+D(t-\hat{t})$

P16. Solution:

$$\hat{x}_{10}=\frac{1}{10}b_{10}+\frac{9}{10}\hat{x}_9$$

P23. Solution:

$$\|P-Q\|^2=\left\|\begin{bmatrix}x\\x\\x\end{bmatrix}-\begin{bmatrix}y\\3y\\-1\end{bmatrix}\right\|^2$$

$$=(x-y)^2+(x-3y)^2+(x+1)^2$$

$$=3x^2-8xy+2y^2+2x+1$$

$$\left.\begin{array}{l}\frac{\partial}{\partial x}\|P-Q\|^2=0 \Rightarrow 6x-8y=-2\\ \frac{\partial}{\partial y}\|P-Q\|^2=0 \Rightarrow -8x+4y=0\end{array}\right\} \Rightarrow \begin{array}{l}x=\frac{1}{5}\\ y=\frac{2}{5}\end{array}$$